

# Complete Project Setup & Deployment Guide

Instructions for setting up the Fusion Hackathon Full-Stack Workspace on a new laptop (Windows, macOS, Linux).

## 1. System Prerequisites

| Tool          | Minimum Version        | Command Check                          | Notes                                   |
|---------------|------------------------|----------------------------------------|-----------------------------------------|
| Git           | v2.30+                 | <code>git --version</code>             | Required for source control             |
| Node.js & npm | v18+ (LTS recommended) | <code>node -v &amp;&amp; npm -v</code> | Runs Vite + React frontend              |
| Python        | v3.12+                 | <code>python3 --version</code>         | Runs FastAPI backend                    |
| uv (Astral)   | Latest (Recommended)   | <code>uv --version</code>              | Blazing fast Python venv & package sync |

## 2. Step 1: Clone the Repository

```
# Open your terminal or PowerShell and clone the repo
git clone https://github.com/UnsettledAverage73/fusion-hackathon.git
cd fusion-hackathon
```

## 3. Step 2: Backend Setup (FastAPI & SQLite)

The backend powers REST endpoints, JWT session security, Google Identity token verification, and real-time Slack alerting.

### A. Copy the Environment Template

```
cd backend
# Linux / macOS:
cp .env.example .env

# Windows (Command Prompt):
copy .env.example .env
```

### B. Install Dependencies and Run

Option 1 (Recommended & Fastest with `uv`):

```
uv sync
uv run uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

Option 2 (Standard Python `venv` + `pip`):

```
# Linux / macOS:
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000

# Windows (PowerShell):
```

```
python -m venv .venv  
.venv\Scripts\Activate.ps1  
pip install -r requirements.txt  
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

**Backend Verification:**

- Health Check: <http://localhost:8000/health> → {"status":"ok"}
- Swagger Docs: <http://localhost:8000/docs>

## 4. Step 3: Frontend Setup (React 19 + Vite + Tailwind CSS)

Open a **second terminal window** and follow these steps:

### A. Navigate & Configure Environment

```
cd fusion-hackathon/frontend

# Linux / macOS:
cp .env.example .env

# Windows (Command Prompt):
copy .env.example .env
```

### B. Install Packages & Start Vite Dev Server

```
# Install node modules
npm install

# Start development server
npm run dev
```

#### Frontend Access:

Open your browser to <http://localhost:5173>. The navbar will display `Render: OnLine` or local connection indicator.

## 5. Free-for-Dev Architecture & Built-in Services

| Service                                     | Implementation Details                                                                                        | Free Tier Limits                              |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Google Identity Services (OAuth 2.0)</b> | GIS button, token verification with Google tokeninfo endpoint, and JWT issuance.                              | 100% Free & Unlimited.                        |
| <b>1-Click Demo Login</b>                   | Direct instant evaluator/judge authentication button in Auth Modal without API keys.                          | Unlimited local / cloud sessions.             |
| <b>Slack Block Kit Notifier</b>             | Ported from <code>Cloud_Cost_Optimization</code> . Standard library HTTP POST to Webhook URL with rich cards. | 100% Free incoming webhooks.                  |
| <b>Render Cloud Deployment</b>              | Frontend static SPA + Backend Python web service automatically tracking Git repo.                             | Free static sites & 750 monthly server hours. |

## 6. Live Hackathon Demonstration Flow

- Sign In Demo:** Click **"Sign In"** in the top navbar → select **"1-Click Judge & Tester Sign In"**. Notice your evaluator profile and role badge appear instantly.
- Slack Live Alert:** Click **"Dispatch Live Slack Alert"** on the main dashboard banner. If a webhook URL is configured in `backend/.env`, a formatted Block Kit alert appears in your Slack channel in real time.
- Interactive Swagger:** Visit `/docs` to show live OpenAPI documentation and demonstrate active REST routes (`/health`, `/api/v1/auth`, `/api/v1/notifications`, `/api/v1/items`).

## 7. Production Cloud URLs

- **Frontend Production (Render):** <https://codeforge-0j8e.onrender.com>
- **Backend API Production (Render):** <https://codeforge-zdxk.onrender.com>
- **Backend Swagger Docs:** <https://codeforge-zdxk.onrender.com/docs>